

● Cloudflare Radar / 2024

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network in 2024.

Audience: Technology executives & network infrastructure leaders

Tone: Professional • Data-driven • Authoritative

Scope: Traffic • Services • Protocols • Disruptions • Security

Global traffic growth (2024)

17% Global Traffic Growth and Starlink's 3.3x Surge

+17%

Global Internet traffic grew year-over-year in 2024.

3.3x

SpaceX Starlink traffic growth during 2024 (normalized to peak request volume).

- **Underlying driver:** increasing reliance on the Internet for communication, commerce, entertainment, and transportation.
- **Method note:** the traffic trend line starts mid-January to normalize for post-holiday activity.
- **Starlink context:** satellite connectivity continues expanding into previously underserved locations (availability not yet global).

Normalized Starlink traffic worldwide showing 3.3x growth in 2024

Figure: Normalized Starlink traffic worldwide (3.3x growth in 2024).

Popular services (Cloudflare 1.1.1.1 DNS)

Google Leads Top 10 Services as Generative AI Rises

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

- **Google remains #1** overall for the second consecutive year.
- **Generative AI continued its rapid rise** throughout 2024, while the Metaverse receded into the background.
- **Methodology:** ranking is based on aggregate data from Cloudflare’s 1.1.1.1 DNS resolver.

Top 10 most popular Internet services worldwide

Figure: Top 10 most popular Internet services worldwide.

● Adoption shifts (2024)

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

2% → 13%

Post-quantum encryption share of TLS 1.3 traffic (early 2024 to peak ~12–13%).

21%

HTTP/3 share of traffic (QUIC-based, faster connections, packet-loss mitigation, encryption by default).

- **Inflection point:** adoption jumped sharply after Google Chrome 124 enabled post-quantum key agreement by default on **April 17, 2024**.
- **Browsers:** Mozilla Firefox began rolling out post-quantum by default; Apple Safari started initial testing.
- **API client languages:** Go is the most popular (≈12% of automated API requests), followed by Node.js, Python (9.6%), Java (7.4%), and .NET (3.6%).

 Post-quantum encrypted traffic worldwide showing rapid adoption after Chrome 124 release

Figure: Post-quantum encrypted traffic worldwide (rapid adoption after Chrome 124).

225 Major Disruptions: Government Shutdowns Lead Outage Causes

225

Major Internet disruptions observed globally in 2024 (Radar Outage Center).

116+

Government-directed disruptions (largest share among identified causes).

Cause	Count
Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1

Causes of major Internet disruptions observed in 2024

Figure: Causes of major Internet disruptions observed in 2024.

● Security & email (2024)

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

Share of all traffic processed by Cloudflare that was mitigated.

4.3%

Share of emails processed that were malicious.

- **Most-attacked sector:** Gambling & Games.
- **Bot concentration:** 68% of global bot traffic originated from the top 10 countries.
- **Log4j persistence:** CVE-2021-44228 remains actively targeted more than two years after disclosure (December 2021).
- **Email threats:** among malicious emails, **43%** contained a deceptive link; TLD **.bar** originated the largest share of malicious and spam email observed.

● Routing security signal

+6.4%

RPKI valid routes increased (improving routing security posture).

Note: Slide highlights only metrics explicitly provided in the source brief.

Five Takeaways Shaping the 2025 Internet Outlook

01

Plan for continued demand growth

Global Internet traffic rose **17%** in 2024; capacity and performance investments remain strategic.

02

Treat post-quantum as a near-term rollout

Post-quantum encryption jumped from ~2% to ~**13%** of TLS 1.3 traffic after **Chrome 124** (Apr 17, 2024).

03

Assume geopolitical fragility in connectivity

Of **225** major disruptions, **government-directed shutdowns (116+)** were the leading cause.

04

Design defenses for persistent, scaled threats

Cloudflare mitigated **6.5%** of processed traffic; Log4j remains actively exploited.

05

Optimize for modern protocols and developer reality

HTTP/3 reached **21%** adoption; Go led API client languages at ~**12%** of automated API requests.

2025 implication: prioritize protocol modernization, crypto agility, outage readiness, and durable mitigation posture.